

Abdullah Al Maruf

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Research Summary

My research examines how vision-language models use visual evidence when scenes are ambiguous, data differ from training conditions, or computational resources are limited. I study these questions through diagnostic benchmarks and controlled experiments, including Bangladesh road-scene hazard evaluation and feature distillation for compact classifiers. Current work extends this focus to Bengali news understanding, relational reasoning in small VLMs, and evidence-based industrial inspection.

Education

Rajshahi University of Engineering & Technology (RUET), Bangladesh

B.Sc. in Computer Science and Engineering

Expected Mar 2027

CGPA: 3.55/4.00 after six semesters

Research Interests

Visual grounding in vision-language models; multimodal evaluation and robustness; efficient visual representations; knowledge distillation.

Publications and Manuscripts

Accepted Workshop Contributions

Abdullah Al Maruf. “DINOv2 Feature Distillation for Recurrent Local-Update Image Classifiers.” *LIGHT Workshop at NeurIPS 2026*. Peer-reviewed extended abstract; accepted poster; non-archival. [\[Paper\]](#)

Abdullah Al Maruf, Md. Sajedul Islam, Tanmoy Mridha, and Irfan Hossain Bhuiyan. “BD-HazardVLM: Probing Vision-Language Models for Latent Defensive-Driving Hazards in Bangladesh Road Scenes.” *ECCV 2026 Workshop on Safe and Defensive Autonomous Driving (SDAD)*. Accepted workshop paper and poster; non-archival. [\[Paper\]](#)

Submitted Manuscript

Abdullah Al Maruf and Tirtho Roy. “Single-Pass Uncertainty for Selective Polyp Segmentation under External Dataset Variation.” *ML4H 2026 Symposium*. Submitted; under review.

Research Experience

BD-HazardVLM

2026

First author; study design and leadership

- Led the design of a 500-image Bangladesh road-scene benchmark covering hazard category, risk timing, severity, defensive actions, and visible evidence.
- The study evaluated three VLMs and found that hazard-presence accuracy of 88.0-89.2% essentially matched the 89% majority baseline; category macro-F1 was 9.4-10.5%.
- Identified failures hidden by aggregate scores: InternVL3-8B achieved 61.2% severity accuracy but missed all 58 high-severity cases.

DINOv2 Feature Distillation

2026

Sole-author study of recurrent local-update image classifiers

- Conducted 12 CIFAR-100 training runs across two architectures, two objectives, and three seeds, with paired initialization, data order, and augmentation views.
- Feature supervision from frozen DINOv2 ViT-S/14 increased recurrent-classifier accuracy from 51.71% to 54.77% (three-seed means); the feed-forward comparator improved from 53.99% to 56.13%.
- Profiled inference costs: the recurrent model required approximately 17.4 times as many MACs as the feed-forward comparator at similar parameter counts.

Research Experience Continued

BanglaNewsGround

Sep 2026 - Present

Undergraduate thesis | Supervisor: Md. Nasif Osman Khansur, Assistant Professor, CSE, RUET

- Comparing journalist captions, text-only rewrites, visually grounded recaptions, and combined supervision to study their effects on Bengali image-text representations.
- Running retrieval and caption-matching experiments with semantic and entity hard negatives to distinguish visual-grounding improvements from language cleanup.

Tiny VLM Relational Reasoning

2026 - Present

Independent research; experiments underway

- Testing question-conditioned structured distillation for sub-billion-parameter VLMs against answer-only, unstructured-rationale, and corrupted-relation controls.
- Evaluating counterfactual pairs and visual-dependence controls to examine whether predictions respond to relation changes and rely on image evidence.

Spec2Inspect-Lite

2026 - Present

Independent research on industrial inspection; core experiments complete

- Evaluated a specification-grounded inspection pipeline on three MVTec AD categories and 60 human-authored specifications, separating image-normalized measurements from deterministic rule execution.
- On 588 matched pairs, false-PASS rate was 7.64% versus 94.44% for schema-constrained Qwen2.5-VL-3B-Instruct, with coverage of 74.66% versus 99.32%. Overall accuracy superiority was not statistically established.
- Diagnosed parsing errors across 7,513 specification-image pairs: verdict accuracy was 82.07% with reference rules and 56.73% with raw Qwen-parsed rules.

Visual Reference Tables

2026 - Present

Collaborative research; manuscript in revision

Ushashi Bhattacharjee, **Abdullah Al Maruf**, Koushik Howlader, Sayantan Chakraborty, and Tirtho Roy.

“Visual Reference Tables Recover Signal That Tissue Captions Alone Miss.”

- Continuing work on whether visual reference tables provide useful information beyond tissue captions.

Technical Skills

Programming	Python, C++
Machine Learning	PyTorch, Hugging Face Transformers, PEFT/LoRA, scikit-learn
Research Tools	Git, Docker, Kaggle
Backend and Data	FastAPI, REST APIs, MySQL, PostgreSQL, SQLite

Honors and Awards

- Champion, University Innovation Hub Program, IC-6 cohort.
- Champion, Hult Prize at RUET, 2025; represented RUET at the national competition.
- 6th place, Inter-University Capture the Flag Competition, BUET CSE FEST, 2024.
- Top-25 finalist, BDApps Innovation Challenge, among more than 1,500 teams nationwide.

Leadership and Engineering

GreenLoop | Founder and CTO

Khoj (Lightning Search)

- Built fuzzy retrieval over 10,000+ entries using n-gram indexing, candidate filtering, and RapidFuzz ranking, exposed through FastAPI.